

NexAI Examination Management System

Complete end-to-end lifecycle flow, multi-portal feature specifications, architectural flowcharts, UML class models, and use case diagrams.

Document	Authority	Technical Scope	Security
Version	Office of Controller of	UML Diagrams, Flowcharts & Core	Classification
v3.4 Production	Examinations (CoE)	Features	Official Academic
Spec			Record

1. Executive Architecture & Role Segregation

The NexAI Examination System enforces strict institutional segregation of duties between **Continuous Internal Evaluation (CIE)** managed locally by Academic Departments, and **Semester End Examinations (SEE)** conducted centrally across all branches by the Controller of Examinations (CoE).

Institutional Jurisdiction Mandate

- Head of Department (HOD):** Manages internal department assessments (CIE 1, 2, 3), 75% student attendance eligibility gates, internal faculty endorsements, and internal hall ticket release.
- Controller of Examinations (CoE):** Exercises autonomous authority over all semester end examinations (SEE), inter-departmental student seating interleaving, question paper vault approval, centralized invigilator duty balancing, and degree ledger publication.

2. Complete Examination Lifecycle Flowchart

The entire examination lifecycle operates as a 5-phase sequential pipeline with cryptographically enforced milestones:

Phase 1: Continuous Assessment & Eligibility Gateway

Faculty & HOD

Faculty marks daily batch attendance (1-click 'Mark All Present' with absentee toggles). Students below 75% are locked at the HOD Eligibility Gateway. Faculty evaluates CIE written tests on-screen. HOD endorses internal marks and generates barcoded hall tickets.



Phase 2: Confidential Question Paper Authoring & Vaulting

Paper Setter & CoE

Internal/External paper setter authors questions with AI assistance, mapping questions to Bloom Taxonomy levels (L1–L6) and module blueprints. Paper is encrypted client-side with AES-256 and committed to decentralized IPFS storage. CoE decrypts and approves multi-set options.



Phase 3: Institutional SEE Examination Scheduling (AI Engine)

CoE Autonomous Solver

CoE specifies multi-department scope, semesters, exams per day, and time slots. AI Constraint Satisfaction engine generates 2D checkerboard student interleaving (zero adjacent same-subject desks) and equalizes invigilator duties ($\sigma \rightarrow 0$) with cross-department supervisor pairing.



Phase 4: Examination Conduct, Collection & Script Digitization

Invigilator & Scanning Officer

Candidate hall tickets verified via door QR scan. Physical booklets collected and delivered to confidential scanning station. High-speed OMR/barcode scanners ingest booklets and mask USN with encrypted fictitious dummy numbers to enforce double-blind evaluation. Scripts bundled into 25-booklet digital packets.



Phase 5: Double-Blind Valuation, Scrutiny & Grade Ledger Publishing

Evaluator, Scrutinizer & CoE

Valuators grade digitized scripts on-screen with automatic arithmetic summing. Chief Scrutinizer runs unvalued-question detection and recalculation audits (scripts with >15% variance routed to third valuation). CoE locks the immutable grade ledger, and students access digital grade cards.

3. AI Seating & Equal Invigilation Engine Flowchart

The mathematical optimization model resolving 2D bench interleaving and faculty workload equalization:



4. Comprehensive UML Class Model (Domain Entities & Methods)

Structural object model capturing primary domain entities, relationships, attributes, and operations:



5. UML Use Case Model (Actor Interactions & Permissions)

Matrix representation of actors, subsystems, and operational use cases across the institutional platform:

Actor	Primary Operational Use Cases	Included / Extended Use Cases
Controller of Examinations (CoE)	• Define SEE Multi-Department Scope & Calendar • Execute AI Seating Interleaving & Invigilator Balancer • Approve Encrypted Question Papers from Vault • Provision Officer Credentials (HOD, Scrutinizer, Scanning) • Lock & Publish Immutable Grade Ledger	<<include>> 2D Anti-Cheating Check <<include>> Duty Equalizer ($\sigma \rightarrow 0$) <<extend>> Emergency Re-allocation
Head of Department (HOD)	• Schedule Department Internal Assessments (CIE) • Enforce 75% Attendance Eligibility Cutoff • Endorse Department Internal Marks • Release Barcoded Student Hall Tickets	<<extend>> Medical Condonation Override <<include>> Fee Clearance Validation
Faculty Instructor	• Record Daily Batch Attendance (Absentee toggle) • Conduct & Grade CIE Internal Written Tests • Annotate Scripts on Digital Correction Studio • Submit CIE Marks to HOD for Endorsement	<<include>> Bloom Taxonomy Rubrics <<include>> CO-PO Attainment Calculation
Confidential Paper Setter	• Generate Questions via AI Assistant • Map Questions to Bloom Taxonomy (L1–L6) • Validate 20-mark Module Choice Blueprint • Encrypt with AES-256 and Commit to IPFS Vault	<<include>> Scheme Compliance Check <<include>> Client-Side Key Generation
Scanning Officer	• Ingest Physical Answer Booklets via Barcode / OMR • Mask USN & Identity with Fictitious Dummy Number • Split & Bundle 25-Script Evaluator Packets	<<include>> Double-Blind Cryptographic Mask <<extend>> Rescan Corrupted Page
Evaluator (Valuator)	• Perform Split-Screen On-Screen Script Valuation • Apply Checkmarks, Crosses, Underlines & Remarks • Enter Question-wise Scores with Automated Summation	<<include>> Anti-Skipping Validation <<include>> Best-of-Choice Calculation
Chief Scrutinizer	• Audit Completed Valuation Packets • Run Automated Unvalued Question & Page Scan • Verify Arithmetic Summation & Section Totals • Escalate Valuation Variance ($>15\%$) to 3rd Valuation	<<extend>> Return to Valuator for Query <<include>> Audit Ledger Logging
Invigilator Proctor	• Scan Candidate Hall Ticket QR Code at Door • Verify Interleaved Seating Compliance • Reconcile Physical Answer Booklets Count	<<extend>> Malpractice Incident Report <<include>> Digital Attendance Lock
Student Candidate	• Download Digitally Signed Hall Ticket with QR • Locate Assigned Hall, Desk & Bench in Real-time • View Continuous Attendance & CIE Score Cards • Access Final Semester End Examination Grade Sheet	<<extend>> Revaluation Application

6. Comprehensive Anti-Malpractice Security Matrix

Vulnerability / Risk	System Safeguard in NexAI	Responsible Portal
Copying from Adjacent Desks	Multi-Department 2D Matrix Interleaved Seating	CoE AI Allocation Engine
Evaluator Favoritism / Bias	Double-Blind Barcode Masking (Fictitious Dummy Numbering)	Scanning Officer & Evaluator
Un-evaluated Questions & Calculation Errors	Digital On-Screen Valuation Studio with Automated Page Audit	Evaluator & Scrutinizer
Unequal Faculty Duty Allocation	Min-Variance Workload Equalizer Algorithm ($\sigma \rightarrow 0$)	CoE Invigilator Balancer
Impersonation & Hall Ticket Fraud	Encrypted QR Code Hall Ticket Verification at Exam Door	HOD Gateway & Mobile App
Question Paper Leaks Prior to Exam	Client-Side AES-256 Encryption & Decentralized IPFS Storage	Paper Setter & CoE Vault